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Metacognitive regulation and ChatGPT-assisted writing: introducing an interactive model for cultivating strategic engagement

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Abstract
Given the contemporary global issues and growth of generative artificial intelligence (GenAI) technology, students are required to effectively activate their metacognitive strategies when using AI not only to develop essential 21st-century skills but also to leverage the benefits of GenAI and avoid its potential negative effect. This paper explores the dynamic realm of AI-assisted writing within the framework of metacognitive regulation (MR) and self-directed learning. The study introduces a data-driven model of MR derived from examining participants' use of MR strategies when employing ChatGPT for academic writing. Adopting an in-depth qualitative approach, semi-structured interviews were conducted with 25 postgraduate students who were purposively selected from three distinct Moroccan universities. Based on reflexive thematic analysis, findings revealed that students with higher MR were found to regulate the use of ChatGPT by aiding them in various aspects, including planning, monitoring, revising, information management, and debugging during their academic writing. Based on the findings, a contextualised metacognitive regulation framework for AI-assisted L3 writing was proposed to explain the interaction between MR and ChatGPT within AI-assisted writing. At the core of this framework is the recognition of MR and self-regulated learning. It posits that, for learners to achieve a higher level of MR, a collaborative effort is required to foster agency and autonomy by effectively leveraging the features and affordances of ChatGPT. The study not only sheds light on the theoretical foundation of MR but also provides empirical findings, offering practical implications for cultivating MR in AI-assisted writing contexts.

Keywords Artificial intelligence, ChatGPT-assisted writing, Academic writing, Metacognitive regulation, Interactive model, Self-directed learning

1 Introduction and theoretical background

In the 21st century, metacognition is widely recognized as an indispensable ability and has long captured the research interest of educational specialists and psychologists [11]. It is the ability that enables students to govern and execute systematic control over their learning, which can lead to enhanced decision-making [7], increased leadership skills [55], and better psychological well-being [21]. Accordingly, the recent educational reforms in Morocco, with its latest plans, namely *the Strategic Vision for Education* 2015–2030 and *ESRI 2030 PACT* [16], advocate for the cultivation of multilingualism [45, 48, 49], autonomous learning, and digital literacy [17]. Now, despite the lack of explicit reference for metacognition in these reform documents, one of their main aims is to create a self-regulated and self-directed learner able to regulate their cognitive thinking and academic learning.

Among the four language skills, academic writing in English is perceived as one of the most daunting and demanding skills, both linguistically and cognitively, especially for non-native speakers and English as a foreign language (EFL) learners [8, 29, 33]. Within the Moroccan context, the writing process becomes ever more complex and challenging for English as a third language (L3) learners due to several factors such as linguistic distance between English (L3) and Arabic (L1) [42], absence of teaching writing as a process [35], and lack of prior training [2]. Other studies have shown that writing difficulties encountered by university students can be attributed to a lack of awareness and use of metacognitive strategies [44]. Hence, existing literature has documented a positive relationship between the use of metacognitive strategies and writing achievement [25] and their significant impact on writing proficiency [24, 46]. Consequently, fostering MR abilities becomes a necessity with the increased attention devoted to higher-order thinking skills in the age of GenAI.

Within the recent emerging technologies and growth of ChatGPT usage, a trend of research has focused on the application of AI in education generally [30, 56] and its potential in enhancing students' proficiency in academic writing, particularly [57, 62], and cultivating critical thinking [26]. Existing research provides inconclusive evidence regarding the impact of AI tools, such as ChatGPT and other Natural language processing models, on writing proficiency within the context of AI-assisted writing [15, 43, 47]. For instance, Alkamel and Alwagieh [3] examined the use of ChatGPT in academic writing among Yemeni learners and reported a significant impact of the tool on their writing accuracy, fluency, and error correction. Similarly, Nugoroho et al. [39] reported that the most frequent uses of ChatGPT were for idea generation and writing enhancement. In addition, Monib et al. [38] documented several benefits of ChatGPT, including personalised feedback and writing assistance, as well as time-saving and accessibility features. On the contrary, other studies raised concerns regarding the limitations and negative side effects that AI could have on students' writing practices, including usage for academic dishonesty [41], reducing creativity [23], and undermining originality and breaching academic integrity [12]. In short, while some studies endorse the integration of these tools to facilitate the writing process, other studies call for careful interaction to maximise the benefits and reduce the negative drawbacks.

To address the aforementioned concerns regarding the use of GenAI, students are not only required to have a higher level of AI literacy but also to effectively implement MR strategies to plan and reflect on their use of AI tools [5, 36]. Existing studies on the interaction between metacognition and AI offer research evidence on the interconnectedness of these constructs. For instance, Teng [52] reported a positive association between metacognitive awareness (MA) and students' practices of using ChatGPT. That is, students with higher MA tend to use ChatGPT actively rather than passively. Consistent findings were reported in studies, such as Yao et al. [57], who indicated that students displayed different MR strategies, such as planning, monitoring, and evaluating, while using ChatGPT. Consistently, Zou and Huang [62] revealed that doctoral students incorporated ChatGPT across different phases of writing, including planning, editing, and revising. While the above studies have shown that AI tools can act as a tool for enhancing metacognitive reflection, Zhang and Zhang [59] found that automated feedback had negative outcomes on students' monitoring strategies. On a similar note, Abdelhalim [1] investigated the MA of learners in utilising ChatGPT to enhance research skills. The study gathered both quantitative and qualitative data from 27 EFL undergraduate students, 12 low MA and 15 high MA, over the course of one semester. The quantitative analysis revealed notable differences in the use of ChatGPT by the low and high MA groups. Additionally, a positive correlation was found between MA scores and students' perceptions of ChatGPT, highlighting varied perceptions and practices between high and low metacognitive learners. In short, most studies highlight the importance of MA in the interaction with AI resources. In other words, students who implement MR strategies at high rates will tend to regulate and establish effective utilisation of AI tools in academic writing.

Based on metacognition theory, MA entails a broad spectrum of metacognitive experiences related to meaning-seeking for goal setting [18]. Brown's [10] model of MA identifies two primary components: Knowledge of cognition and regulation of cognition. The former involves understanding one's own cognitive processes and strategies [14], while the regulation of cognition entails employing MR strategies such as planning, monitoring, and evaluating to control cognitive and learning activities [19]. This MR was further elaborated on to include five key subprocesses, namely (1) planning, which involves establishing objectives and effectively distributing resources, (2) monitoring which refers to the ongoing control and regulation during the learning experience (3) evaluating, which involves critically assessing both performance outcomes and the effectiveness of the strategies used (4) information management strategies, which entail skilfully handling information through techniques such as organising, summarising, elaborating and concentrating, (5) debugging strategies, which focus on correcting misunderstandings and errors in performance and learning [50]. Collectively, these strategies provide a theoretical basis for examining MR to improve learning and performance.

Existing metacognitive models fall short of addressing how humans think, act, and interact within digitally enhanced learning contexts [34]. Although previous metacognitive frameworks were originally proposed to investigate metacognitive strategies in traditional educational settings, with some adaptations to digital environments, the lack of sufficient research evidence prompted the current study to propose a contextualised conceptual framework to adapt MR strategies in AI-assisted writing. Despite the existence of recent frameworks devoted to AI and metacognition in digital environments such as the integrated framework for metacognition in virtual reality (VR) by Teng [51] and Metacognitive Resource Use (MRU) framework by Mizumoto [37], these frameworks only focused on situating learner autonomy and metacognition within different digital environments, while no explicit mention of writing competency was present in both models. More specifically, there is currently no conceptual framework specifically tailored to the context of L2/L3 writing that integrates MR processes while addressing psychological mechanisms in AI-assisted writing. Hence, the study not only offers empirical findings but also introduces a contextualized conceptual model that extends existing metacognitive frameworks to the AI-assisted writing context.

While surveying national and international literature on the interaction between metacognition and ChatGPT in an AI-assisted writing context, it was evident that there is a very limited number of studies examining students' use of MR strategies in ChatGPT for academic writing [52, 53, 57]. Most of this research was only conducted in an L2 writing context. Thus, there is a dearth of studies pertaining to the application of MR in using ChatGPT for L3 writing. So far, no study has been conducted on these constructs within the Moroccan educational context. When it comes to the intersection between metacognition, AI, and writing, there is currently no specific conceptual framework dedicated to evaluating MR while interacting with AI, neither in the L2 writing context nor in L3 settings.

Overall, research on metacognitive engagement with GenAI in academic writing remains in its infancy. Therefore, with the limited number of studies at the international sphere and absence of studies at the national level, the current study sets to fill this critical gap by particularly examining students' usage of MR strategies when using ChatGPT for English (L3) writing, exploring perceived benefits and challenges of ChatGPT-assisted writing, and introducing a new conceptual framework for the interaction between MR and ChatGPT in an AI-assisted writing context. To achieve these objectives, three primary research questions are addressed:

RQ1. How do Moroccan post-graduate students make use of various MR strategies (i.e., planning, monitoring, evaluating, information management, and debugging) in the application of ChatGPT in L3 writing?

RQ2. What are post-graduate students' perceptions of both the benefits and limitations of ChatGPT in navigating their L3 academic writing?

RQ3. What conceptual model can represent learners' MR when interacting with AI tools in an AI-assisted writing context?

Through examining how MR processes, including planning, monitoring, and evaluating, information management, and debugging, affect the use of ChatGPT within an AI-assisted context, the study seeks to provide implications for effective instructional practices that leverage MR to support and regulate student use of AI in academic writing. Additionally, based on the outcomes obtained, this study will also contribute to theoretical discussions by proposing a contextualized theoretical model for the integration of MR in an AI-assisted writing context.

2 Methodology

2.1 Participants and context

The target population of the current study was EFL master's students majoring in various fields, including pragmatics, ELT, and applied linguistics, among others. Participants were drawn from three distinct Moroccan institutions, including Ibn Zohr ($N = 10$), Chouaib Doukkali ($N = 11$), and Hassan 1st ($N = 4$) Universities. Invitations were sent to 32 students, and 25 of them responded and volunteered to participate in this research. 25 post-graduate students, including 9 males and 16 females, were purposively selected due to their experience in using ChatGPT for academic writing, English language proficiency, accessibility, and willingness to engage in the study. The participants' age ranges from 20 to 50 years old, with a mean age of 24 ($M = 24.72$, $SD = 6.079$). They were enrolled in diverse master's programmes within the field of humanities, including applied linguistics and language teaching, cross-cultural communication and visual media, and intercultural studies. Although some participants were from the Agadir region (speaking Tamazight), all of them were L3 English learners, verified via demographic interview questions.

Each participant was identified by unique pseudonyms composed of two letters based on their first and last name to maintain strict confidentiality and anonymity. All participants had a minimum of two years of experience in actively using a variety of AI tools, such as natural language processors (ChatGPT, copilot, Gemini) and writing assistants (Grammarly). All participants were classified as Advanced on Rosetta Stone, based on the platform's internal descriptors corresponding to CEFR level B2 + or above. Note that this language learning tool is a standardised platform used across all Moroccan universities for language proficiency testing. Detailed demographic information of the participants is exhibited in Table 1.

2.2 Data collection and procedures

To obtain a deeper understanding of students' utilisation of MR strategies while using ChatGPT in writing, twenty-five participants were interviewed to collect delicate insights. According to Guest et al. [22], data saturation can happen with as few as 12 participants in homogeneous groups. In alignment with reflexive thematic analysis, we assessed the adequacy of our dataset based on analytic sufficiency rather than statistical saturation. Analytical sufficiency refers to the point at which further data collection no longer yields new, conceptually meaningful insights. This was determined through iterative coding and comparison of transcripts, supporting the sufficiency and representativeness of the sample.

By approximately the 20th interview, subsequent data largely reinforced established meanings and thematic aspects, highlighting that the dataset was sufficiently rich and comprehensive to support robust analysis. The semi-structured interview was designed and informed by numerous scholarly works, including Yao et al. [57], Teng [52, 53], and Kim et al. [31]. The interview was divided into two parts; part one contained four demographic questions, while the second part was composed mainly of thirteen questions spanning across the five dimensions of metacognition, which include planning (3 questions), monitoring (2 questions), evaluating (4 questions), information management (2 questions), and debugging (2 questions). The complete interview protocol is provided in Appendix A.

After carefully designing the interview guidelines, the interviews were conducted at the participants' approved dates and times. Before embarking on the interview, we explained the study's objectives, outlined the interview structure, and ensured the transparency and objectivity regarding the confidentiality of the data. The interview was sequenced following the order of the metacognitive dimensions, which provided smooth navigation between the different components of the interview.

These interviews were conducted via online phone calls via WhatsApp in English and audio recorded with the consent of the informants between January and March 2025. Each interview lasted between 30 and 40 min, with a mean of 35 min. Interviews were conducted in English, the participants´ third language (L3), to allow them to express

Table 1 Demographic information of participating students

Pseudonym	Gender	Age	Master program	University	Rosetta stone scores
OA	Male	30	Applied linguistics and Language Teaching	Ibn Zohr (IBZ)	Advanced (B2+/C1)
ZG	Male	20	Applied linguistics and Language Teaching	IBZ	Advanced (B2+/C1)
MJ	Male	23	Applied linguistics and Language Teaching	IBZ	Advanced (B2+/C1)
LJ	Male	50	Cross-Cultural Communication and Visual Media	Chaouib Doukkali (UCD)	Advanced (B2+/C1)
AM	Male	21	Applied linguistics and Language Teaching	IBZ	Advanced (B2+/C1)
WJ	Male	23	Applied linguistics and Language Teaching	IBZ	Advanced (B2+/C1)
AL	Male	24	Applied linguistics and Language Teaching	IBZ	Advanced (B2+/C1)
AD	Male	30	Cross-Cultural Communication and Visual Media	UCD	Advanced (B2+/C1)
AB	Male	23	Applied linguistics and Language Teaching	IBZ	Advanced (B2+/C1)
KA	Female	23	Applied linguistics and Language Teaching	IBZ	Advanced (B2+/C1)
SA	Female	23	Applied linguistics and Language Teaching	IBZ	Advanced (B2+/C1)
HJ	Female	22	Cross-Cultural Communication and Visual Media	UCD	Advanced (B2+/C1)
KO	Female	22	Cross-Cultural Communication and Visual Media	UCD	Advanced (B2+/C1)
FZ	Female	26	Cross-Cultural Communication and Visual Media	UCD	Advanced (B2+/C1)
NO	Female	23	Cross-Cultural Communication and Visual Media	UCD	Advanced (B2+/C1)
CM	Female	22	Cross-Cultural Communication and Visual Media	UCD	Advanced (B2+/C1)
IC	Female	22	Cross-Cultural Communication and Visual Media	UCD	Advanced (B2+/C1)
GM	Female	22	Cross-Cultural Communication and Visual Media	UCD	Advanced (B2+/C1)
SE	Female	32	Cross-Cultural Communication and Visual Media	UCD	Advanced (B2+/C1)
WL	Female	21	Cross-Cultural Communication and Visual Media	UCD	Advanced (B2+/C1)
OB	Female	21	Intercultural studies	Hassan 1st	Advanced (B2+/C1)
BJ	Female	22	Intercultural studies	Hassan 1st	Advanced (B2+/C1)
WI	Female	23	Intercultural studies	Hassan 1st	Advanced (B2+/C1)
YB	Female	27	Applied linguistics and Language Teaching	IBZ	Advanced (B2+/C1)
CB	Female	23	Intercultural studies	Hassan 1st	Advanced (B2+/C1)

their experiences and strategies in their L3 writing and demonstrate their communicative proficiency, given that they were all English majors.

2.3 Data analysis

Subsequently, the audio recordings were uploaded to the Word software for transcription purposes. Afterwards, they were transcribed using the transcribe feature, which conducts an automatic speech recognition (ASR) to convert the recordings into text.

Subsequently, the interviews were reviewed and cross-checked for accuracy purposes before conducting the thematic analysis. It is worth noting that we only deleted instances of repetition from the transcripts while preserving the conversation's genuineness. Thus, it is worth noting that the grammatical mistakes present in the quotes were produced by the informants.

Adopting reflexive thematic analysis enabled us to identify, analyse, and report patterns of meaning within the interview data [9]. Reflexive thematic analysis aligns with an interpretivist epistemology, which recognises researcher subjectivity and reflexivity as integral to knowledge construction rather than sources of bias. Accordingly, our analytical approach was interpretive rather than reliability-driven; thus, inter-coder reliability indices were not calculated. Instead, meaning-making was understood as a collaborative and reflexive process.

More specifically, the researchers employed a hybrid deductive–inductive approach, moving iteratively between close engagement with the data and theoretical underpinnings. The analysis involved repeatedly reading the transcripts, generating initial codes, and developing themes and sub-themes through a recursive process. The initial coding framework was informed by key metacognitive dimensions derived from theory and mainstream literature [54, 60]. Accordingly, five prominent dimensions of metacognitive regulation (MR) strategies guided the deductive phase of analysis: planning, monitoring, evaluating, information management, and debugging strategies.

At the outset, a deductive coding scheme based on these five metacognitive constructs was developed. As the analysis progressed, additional inductive codes emerged to capture unanticipated dimensions of participants' reflections, particularly those related to their metacognitive interactions with ChatGPT in academic writing. The coding framework evolved iteratively through cycles of reading, coding, documentation, and reflective discussion. Given the hybrid nature of the thematic analysis and the situated context of GenAI-assisted writing in which MR operates, the coding process remained flexible and inclusive throughout.

Subsequently, the analysis shifted towards a more inductive orientation to accommodate newly emerging themes and sub-themes identified across the transcripts. As detailed in the results section, several novel themes were generated, including regulating ChatGPT's performance and learning to craft and refine prompts, which were situated within the broader dimensions of monitoring and planning, respectively.

Regarding analytic rigour, and consistent with the principles of reflexive thematic analysis, the researchers conducted manual coding independently. Each coder reviewed the data individually to identify salient patterns before engaging in collaborative analytic discussions to compare interpretations and co-construct themes. This process was not intended to achieve inter-coder consensus but to promote reflexive engagement and deepen interpretive insight through multiple perspectives. Several sub-themes were refined and aligned with the core features of MR strategies within the adopted framework. Full details of the coding scheme, including themes, sub-themes, and illustrative quotations, are provided in the supplementary materials.

2.4 Reflexivity and positionality

All three analysts have backgrounds in L3 writing pedagogy and research. Two have incorporated GenAI tools in language teaching and research, while one has critically approached the process as a researcher of ICT (Information and Communication Technology). These positionalities have shaped and informed our interpretations. Throughout the analysis, we negotiated our positional perspectives to ameliorate reflexive awareness.

3 Results

To answer the first research question regarding students' implementation of MR strategies while using ChatGPT in L3 writing, the thematic analysis revealed diverse strategies spanning across the five metacognitive dimensions. The results are divided into two major sections. The first one is related to each of the five dimensions of MR strategies used, while the second one is devoted to the perceptions behind using this technological tool. The results are presented in accordance with the sequence of the research questions to ensure clarity and coherence.

3.1 Participants' utilization of MR strategies in ChatGPT for English academic writing

Two main themes emerged within the planning strategy. Concerning the first theme, establishing goals in the application of ChatGPT for writing, three sub-themes emerged, encompassing proofreading and correcting grammatical errors, brainstorming and collecting ideas, and generating essay models and incorporating the output for relevant writing. The majority of informants (24 out of 25) indicated that the first goal behind using ChatGPT was to gather ideas and polish their writing. For instance, AL claimed that he would use ChatGPT for brainstorming and collecting ideas in the process of forming a unique piece of writing. KZ joined the voice of the latter participant by asking ChatGPT for elaborations, while maintaining originality at the same time. All the participants claimed that they would use the tool for editing, proofreading, and enhancing the style of their writings. As KA noted,

My goals would be to have a good piece of writing, good punctuation, good grammar, of course, and I would also use ChatGPT for like advanced type of vocabulary, formal, or maybe even professional, just to make the piece of writing, to some extent, like intellectual (interview response).

On the contrary, SM refrained from the majority through claiming that he would not use ChatGPT for making outlines or generating ideas. As SM asserted,

I wouldn't rely on it to create an outline or even write the essay because it cannot read my ideas. The flow of my ideas is not the same as ChatGPT's. How I see things, how I see the world, it cannot be translated. Therefore, I should rely on myself to write and to translate my ideas….I wouldn't trust GPT to do that for me.

Furthermore, more than half of the interviewees (13 out of 25) reported that they use ChatGPT to provide an essay model, which helps them become familiar with the structure and characteristics of an essay. Regarding the second theme, *developing strategies to accomplish the writing task*, two sub-themes emerged: *learning and offering detailed prompts* and *using ChatGPT to refine language*. On one hand, most students (23 out of 25) indicated that they actively use ChatGPT to revise the language of their writing by first completing the essay and then submitting it to ChatGPT for refinement (sub-theme: *using ChatGPT to refine language*). As HJ noted, "I usually use it to just refine my essays that I am required to submit. I do write the essay, and I send it to ChatGPT and ask for feedback, areas for improvement for my paragraphs, the mistakes that I have made, and how I can improve them." On the other hand, all students demonstrated awareness of the importance of crafting appropriate prompts to elicit optimal responses. For instance, AB stated,

> *I have been experimenting with prompts, trying different ones to see which are more effective and which are not, so I believe I am, to some extent, aware of how I can make my use of the tool more effective.*

The above excerpt suggests that training on how to formulate prompts effectively can lead to more optimal use of ChatGPT in academic writing.

3.1.1 Monitoring

With regard to the second metacognitive dimension, monitoring, one major theme emerged: *monitoring the progress of the writing task*. This theme encompassed two main sub-themes: *regulating the performance of ChatGPT* and *avoiding plagiarism while ensuring originality*. Students monitored ChatGPT's output by carefully examining the generated content for flaws and errors. For instance, KO stated: "I reread; there would be a few mistakes, and I would go through and correct errors, things that maybe I didn't ask for or ChatGPT got wrong from my writing." In other words, KO engages in an evaluative reading of the content to make amendments and remove erroneous information. Additionally, eight other participants reported that they emulate the writing structures produced by ChatGPT as a strategy for self-improvement.

I always use it in order to improve my writing, and I always review the version that the AI gave me and try to acquire skills like structure and to imitate them in subsequent pieces of writing (KO).

Through carefully examining and monitoring the essays produced by ChatGPT, the above participant managed to acquire the structures for future use in other writing tasks. Another notable issue was raised through students having a high level of awareness with regard to risks of plagiarism that can be caused by using ChatGPT. All the participants, unanimously, have been mindful and careful in using this GenAI tool. As one of the participants, AM, asserted his caution by stating *I am very cautious, especially because I'm writing my thesis, so I try to avoid copying or pasting... I always leave distance between the output of ChatGPT and what I would write....just try to paraphrase and put it differently*. The latter excerpt illustrated the avoidance that most participants have when it comes to ChatGPT and plagiarism. They claimed that using the content without modification would lead to plagiarism detection. As a result, students monitored ChatGPT by using paraphrasing strategies and learning structural properties of the tool to avoid academic dishonesty and maintain authenticity.

3.1.2 Evaluating

The dimension of evaluation yielded an array of themes and subthemes. This can be attributed to the user experience of more than two years that students had with ChatGPT and other GenAI tools. All the informants were capable of spotting the strengths and weaknesses of applying this tool in their English academic writing. The first identified theme was rethinking the roles of the human writer and ChatGPT. The latter generated one sub-theme, namely: Human touch is a necessity. All the interviewees claimed that every individual should be responsible throughout the writing process, whereas ChatGPT can serve as a proofreader for addressing language issues in their writings.

It is important to remember that intelligence is a human attribute, and because it's a human attribute, we do not have to feel like this program can outsmart us. It's just a way of research in the sense that it can help us synthesize what we know. So the way we ask the question is the way it gives it. It doesn't go out of what we ask (SE).

The excerpt highlights the distinction between human intelligence and artificial intelligence. Participant SE stated that AI cannot replace or surpass human abilities, emphasizing that humans remain the ultimate authority in guiding and using AI tools responsibly. Supporting this view, AD remarked: *"After using ChatGPT for a long time… it is a robot, it is AI. The language itself is artificial. So the product is not as good as, or as authentic or natural, as it would be if I wrote it myself."* In other words, human writing demonstrates originality and authenticity, which AI-generated content cannot fully replicate. AI writing, therefore, cannot achieve the same level of accuracy or authenticity as human writing.

The informants also identified several strengths of ChatGPT, such as efficiency, time-saving, content expansion, and assistance with editing structure and refining language. However, they noted some weaknesses, including its tendency to hinder creativity, produce a robotic style, generate unexpected responses, and provide unreliable information. Notably, most students ($N = 15$) considered ChatGPT beneficial for improving structure and expanding content. For instance, CB explained: *"I make mistakes, like forgetting the 's' with the third person singular… I just use ChatGPT to fix that."*

Honestly, there are so many benefits. My vocabulary first. Because it gives you plenty of expressions that can be used frequently in pieces of writing. Also, it provides so many synonyms…. After using… I think I could tell that my writing skills were really enhanced in terms of structure and vocabulary enrichment (WL).

While ChatGPT has played beneficial functions by assisting students in expanding ideas, proofreading, enriching vocabulary, and writing style, students have also uncovered several deficiencies pertaining to this technological tool. ChatGPT was deemed disadvantageous due to its non-humanistic style and untrustworthy data. Overall, complex perceptions were evoked pertaining to the use of ChatGPT in academic writing. Rather than blindly over-relying and trusting the tool, students demonstrated a heightened level of caution and awareness based on their user experience. They identified fruitful strengths and areas of improvement to maximise the benefits of this technological tool in academic writing.

3.1.3 Information management

Three themes were identified within the aspect of information management. Regarding the first theme, which involves organising and summarising information for the writing task, students exhibited variations in their use of ChatGPT. Some students refused to utilise ChatGPT to process the information, primarily due to their belief that ChatGPT is biased and inaccurate. For example, *'… the tool hallucinates sometimes when I ask a certain thing, it creates ideas and information to satisfy my need, not necessarily to report a true fact'* (AB, interview response). However, due to the daunting task of reading extensive materials weekly, some students relied on ChatGPT to help them process the information by summarising articles and books. For instance, OB employed ChatGPT to summarise the ideas of the literature *by reporting in the interview that 'I use it for summarising text readings instead of wasting a lot of time in reading a whole chapter. ChatGPT provides me with summaries or detailed summaries*. Others used ChatGPT to process difficult concepts. As BL described his use of ChatGPT as looking for *'Some pieces of information that I need to know. I go up, and I will ask ChatGPT to tell me, what does this mean? Like if it were about notions like language acquisition. I will ask the tool Can you tell me what language acquisition is for collecting information'.*

Meanwhile, as reported in the previous sections, the students recognised that some information generated by ChatGPT was unreliable and inaccurate. Therefore, they continuously inspected the credibility of the information generated by ChatGPT, which emerged as a second theme within the information management dimension in the context of AI-assisted learning. After retrieving a reading text along with questions, about a famous Moroccan inventor, *Rachid Yazami*, from ChatGPT, AM noticed, shockingly, that ChatGPT offered among the questions when the inventor died. However, this was a totally invalid statement since the inventor is still alive. This incident contributed to erroneous information, which increases the untrustworthiness of ChatGPT.

3.1.4 Debugging

Within the dimension of debugging, students' strategies focused on correcting errors in the information generated by ChatGPT. Two subthemes emerged: *refining prompts and referring to additional tools*. When students identified errors in ChatGPT's output, the most frequently mentioned strategy was to refine their prompts and make follow-up inquiries. As HJ explained: *"I first run my eyes over what it (ChatGPT) gives me, and then I try to see the areas that it did not tackle appropriately as I expect, and I try to give examples and paraphrase my prompts."* Overall, when students encountered a mismatch between their expectations and the output generated, they either iteratively revised their prompts or cross-checked the information using other resources, the latter forming the second subtheme.

In addition, students utilised supplementary tools and resources, such as Google or academic databases, to verify the information provided by ChatGPT. One participant noted: *"I usually ask ChatGPT to provide me with the source, and then I go to the original source and check it myself to make sure that the information is accurate."* Others relied on Google Scholar to locate research articles, thereby ensuring the accuracy and reliability of ChatGPT's output.

3.2 The participants' perceptions of ChatGPT in academic writing

The findings revealed that postgraduate students held positive attitudes toward the integration of ChatGPT in academic writing. All participants unanimously acknowledged the transformative impact of ChatGPT on their writing. They reported notable improvements in vocabulary expansion, idea elaboration, and overall writing style. One participant even noted that they planned to incorporate ChatGPT into their classes, citing its positive effects on students' development across the four language skills.

Despite their advocacy for the tool, students consistently emphasized the importance of using ChatGPT appropriately and critically to achieve desired outcomes. Rather than over-relying on the technology, participants stressed that the tool should serve as an

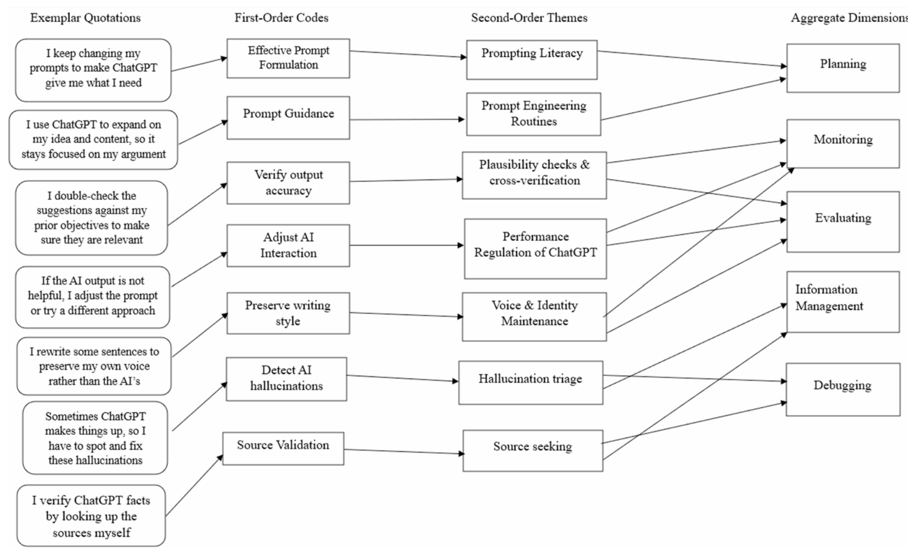

Fig. 1 Adapted metacognitive regulation framework for AI-assisted L3 writing. Arrows represent the analytic progression from exemplar quotations to first-order codes, second-order AI-specific sub-mechanisms, and aggregate MR components

assistant and peer, providing feedback and enhancing writing quality, rather than performing the work entirely on the user's behalf.

3.3 An interactive framework for understanding MR in ChatGPT-assisted writing

Several key patterns emerged from the data, including three key observations. First, all the informants reported using three or more of the five MR strategies, suggesting that learners do not passively rely on ChatGPT but engage in a strategic approach while interacting with the tool. Secondly, monitoring and evaluating were the most frequently reported strategies used, implying that students exerted cognitive control over ChatGPT to enhance understanding and assess information. Last but not least, disparities emerged at the level of debugging strategies. That is, students with more experience in using AI tools exhibited more polished debugging strategies, such as knowledge of prompt engineering and experimentation with different prompts.

To build on the thematic findings from the participants' accounts and showcase the interplay between learners´ metacognitive engagement and their AI-assisted writing practices, an interactive conceptual model of MR in AI-assisted writing was developed (Fig. 1). This model illuminates how students engage in MR while using ChatGPT for writing tasks and how their decision-making is influenced by these interactions. The model builds on the established dimensions of MR processes, while incorporating several AI-specific enactments that emerged inductively from the participants´ practices. These AI practices represent situated adaptive strategies through which learners governed their interaction with ChatGPT during academic writing tasks.

Consistent with a reflexive thematic analysis approach, themes were identified and refined based on their interpretive significance and coherence across participants' accounts, rather than on their numerical frequency. This ensured that the analytic focus remained on the meaningful patterns of experience and perception within the data. Figure 1 presents the data structure following Gioia [20], linking exemplar quotations, first-order codes, second-order themes, and aggregate dimensions.

Figure 1, presents a data structure following Gioia [20], illustrating the development of the interactive model of MR in AI-assisted writing. More specifically, Participants´ practices (first-order codes) were interpreted as AI-specific sub-components (second-order themes) aligned with the core five metacognitive components (Aggregate dimensions). Within planning, participants demonstrated prompting literacy and prompt engineering routines, suggesting intentional experimentation with input phrasing and task organising. Within the monitoring and evaluating dimensions, which contained plausibility checks, cross-verification, and performance regulation of ChatGPT, participants critically evaluated AI outputs for accuracy, relevance, and stylistic appropriateness. These two metacognitive dimensions, monitoring and evaluating, also included voice and identity maintenance, whereby participants actively edited and reframed AI-generated text to maintain personal authorship and authentic rhetorical style. Finally, information management and debugging contained hallucination triage and source seeking, representing careful and cautious handling of misinformation and source credibility .

Central to this framework is the emphasis on the crucial role of MR in fostering learner agency, academic writing, and vice versa. This relationship between MR and ChatGPT for writing is pivotal, as it underlines the capacity of learners/writers to monitor, regulate, and effectively navigate their writing experiences within ChatGPT-assisted writing contexts.

Thus, cultivating MR in a ChatGPT-assisted writing environment, learners would not only benefit from increased self-regulation and MA but also increased writing proficiency and a more self-directed use of the technological tool. Regardless of the model tackling ChatGPT as the main technological tool, it should be noted that this framework can be applicable to other Gen-AI tools within the AI-assisted writing context. The claims made in this model are not addressed to encourage learners to over-rely or depend heavily on the tool, but to provide them with strategies needed to effectively interact with the tool and use it as an assistant and facilitator more than the mere performer of the task. Hence, the main claim that can be retained from this model is as follows: an increased MR of ChatGPT was associated with a more productive AI-assisted writing process.

To navigate and harness the transformative potential of ChatGPT in an AI-assisted writing context, this paper proposes a new model, namely, metacognitively regulating ChatGPT in a writing context, designed to articulate the role of MR in learning/writing development and to maximise the usage of current technologies for enhancing learning. It presents an innovative and promising educational and research ground that harmoniously merge the dimensions of MR of planning, monitoring, evaluating, information management, debugging, along with technological tool such as ChatGPT, into an interactive, reflective learning experience.

Overall, the model visualises how learners' metacognitive strategies intersect with their emerging forms of AI literacy. It posits that while the broad five metacognitive dimensions mirror those found in prior L2/L3 writing research, the novel contribution of this study lies in identifying AI-specific sub-mechanisms that shape how writers plan, monitor, and refine their interaction with GenAI tools. This interactive framework offers a conceptual basis for comprehending how metacognitive regulation is being adapted in the context of AI-assisted academic writing.

4 Discussion

The findings uncovered a range of MR strategies employed by postgraduate students when using ChatGPT for L3 English academic writing. Within the context of generative AI-assisted writing, these MR strategies reflected patterns similar to those observed in other educational settings. While each metacognitive strategy served a distinct function, a common theme was the interconnectedness of the various MR components. The discussion of these findings in relation to the literature is structured around the five metacognitive components, mirroring the organization of the results section.

4.1 Planning: objectives, application, and resources

Under the dimension of planning, students clearly articulated their objectives for using ChatGPT, which included proofreading and correcting errors, brainstorming and generating ideas, and enhancing their writing style. This finding aligns with those of Črček and Patekar [13] and Levine et al. [32], whose participants primarily used ChatGPT for idea generation prior to beginning the writing process. After completing their essays, students employed the tool to improve various linguistic and mechanical aspects of their writing—a pattern supported by studies such as Allen and Mizumoto [4], which reported that students frequently used ChatGPT for language editing purposes.

Furthermore, participants in our study sought to elevate their writing style to sound more intellectually sophisticated by analyzing ChatGPT-generated model essays and using them to better meet the conventions of academic writing. These findings are consistent with those of Alkamel and Alwagieh [3] and Yao et al. [57], who found that ChatGPT was effective in enhancing overall writing quality, particularly in terms of fluency and accuracy.

Importantly, our findings highlighted that students employed iterative prompting techniques to achieve their desired outcomes. This high level of awareness regarding the critical role of prompting appeared to stem from prior user experience, as well as from some participants having received previous training in prompt design. Similar observations were made by Huang et al. [28] and Monib et al. [38], who demonstrated the importance of integrating prompting strategies to enhance students' interactions with ChatGPT and to facilitate personalized feedback. Overall, clearly establishing goals and objectives for using ChatGPT enables students to approach the tool in a structured manner and to identify their own needs and weaknesses before engaging in writing tasks or other academic activities.

4.2 Monitoring: performance regulation and originality maintenance

Despite ChatGPT showing promising and positive effects on students' writing skills, AI remains a concern when it comes to issues related to academic integrity and ethics, as attested in several studies [6, 12, 41]. Hence, the participants in this study continuously tracked the output generated by ChatGPT. Given the issues that arise from using this tool, particularly due to academic dishonesty, all the participants have been mindful in their use of the tool for academic writing through employing monitoring strategies such as checking the accuracy and surveying the structure of the content produced. Due to some errors generated by the tool, students would continuously monitor the output provided for further enhancement and appropriateness.

Moreover, students expressed concerns about the widespread use of ChatGPT for academic and dissertation writing, noting the potential for academic misconduct, particularly plagiarism. Consequently, they emphasised the importance of maintaining originality and authenticity in their work. To this end, participants made minor revisions to ChatGPT's output to preserve the novelty of their writing and master's theses. These findings are consistent with those of Huang et al. [27], who reported that students with stronger self-regulatory skills use generative AI tools to enhance creativity while minimising the risk of plagiarism.

4.3 Evaluating: strengths and weaknesses of ChatGPT

Several strengths and weaknesses of using ChatGPT for academic writing emerged from the study. Participants distinguished between ChatGPT as an assistant tool and as a replacement for human effort. Specifically, students recognized the utility of AI in supporting various aspects of writing while emphasizing the essential role of human effort in managing and conducting the writing process. Accordingly, students wrote their essays themselves, viewing ChatGPT primarily as a provider of feedback rather than as a task-doer. This distinction highlights students' active metacognitive knowledge and regulation, as they navigated different phases of the writing process while leveraging ChatGPT for linguistic refinement [57, 62]. This finding aligns with Teng [53], who emphasized that metacognitive awareness enables students to use ChatGPT responsibly as a support tool rather than a substitute for their own thinking and writing.

Conversely, participants also reported several weaknesses of ChatGPT in academic writing, including a robotic writing style, unexpected or misinterpreted responses, unreliable information, and the potential to hinder creativity if over-relied upon. These limitations are consistent with previous studies [23, 39, 58]. Overall, students' interactions with ChatGPT outputs not only stimulated the practice of evaluation skills but also enhanced their reflective and metacognitive regulation (MR) abilities. These findings mirror those of Oates and Johnson [40], whose participants demonstrated improved critical evaluation skills through engagement with ChatGPT-generated essays.

4.4 Information management: navigating the trustworthiness of the information

When it comes to information processing, students employed ChatGPT to conduct a summary of different content and seek conceptual knowledge to be incorporated in their essays. As the participants are required to carry out weekly readings and summaries of chapters, they actively resorted to the technological tool to ease and save time from reading the whole content. However, students continuously questioned the reliability of the information produced by ChatGPT. Despite students reporting that ChatGPT can produce unreliable information, they selectively picked the information to be incorporated into their writings. But despite this vigilant and cautious management of information generated by ChatGPT, over-dependence on the tool for information processing and source generating may pose undesirable effects on students' critical and creative thinking skills [26]. Nevertheless, all the participants consulted other resources to mitigate the weaknesses of ChatGPT in this department, as one of the participants, WL, asserted that *ChatGPT should not be the main source to process information*. This mitigation served as a theme under the debugging dimension.

4.5 Debugging: mitigating erroneous information by different means

Under this dimension, students were observed to mitigate the errors produced by ChatGPT by employing various debugging strategies, including refining prompts and using supplementary tools for cross-checking. In the established literature on metacognition and writing, such strategies are typically reported as focusing on correcting structural or informational issues within the written product itself. However, in the context of AI-assisted writing, these strategies were applied to address the mistakes generated by ChatGPT. The findings indicate that students continuously adjusted their prompts to achieve desired outcomes and correct misconceptions introduced by the AI. Implementing these debugging strategies not only helped students manage the limitations of ChatGPT but also enhanced their reflective thinking and deepened their understanding of the potential pitfalls associated with this technological tool.

4.6 Interpretation of the interactive model

The data-driven conceptual model developed in this study highlights how learners interact with ChatGPT to support their writing and MR processes. More specifically, the model illustrates the interactive relationship between students' strategic use of ChatGPT, their metacognitive regulatory strategies, and their perceptions of AI-assisted writing. The model integrated the five key metacognitive processes: planning, monitoring, evaluating, information management, and debugging with several AI-specific enactments, including prompting literacy, prompt engineering, performance regulation, plausibility checks, voice and identity maintenance, hallucination triage, and source seeking. These metacognitive strategies and AI mechanisms contribute to a dynamic, self-directed process in which ChatGPT serves both a cognitive assistant and a metacognitive cultivator. Using a Gioia style structure, the mode visually maps these interactions among the two main constructs, AI-assisted writing and metacognitive regulation. In this way, it captures both the interpretive depth of learners' self-reported practices and the reflective and interactive process of AI-assisted writing that can stimulate students' constant MR.

This interactive model provides several practical implementations for the integration of MR and GenAI tools like ChatGPT in technology-enhanced writing contexts within educational environments. It probed key aspects for supporting MR in AI-assisted writing for educators, curriculum and technology designers, and researchers:

- *Metacognitive-instructional support*: educators can design courses that harness each metacognitive component within AI-assisted writing contexts (e.g., teaching students prompting strategies before using ChatGPT for writing or how to check and assess their understanding and writing with ChatGPT).
- *Cultivation of AI use coupled with metacognitive prompts*: Instructors can raise students' awareness regarding the utility of Gen-AI tools through encouraging them to use ChatGPT not just for generating content, but as a tool for self-reflection and questioning, synthesis, and error-checking. Educators can train students how to systematically govern interactions with AI, monitor and evaluate ChatGPT output, and how to amend inaccurate output and adjust prompts.
- *AI-enhanced writing programs*: Curriculum designers can design writing activities and tasks that incorporate the five metacognitive processes. For example, assignments that can include reflective journals that stimulate students' thinking

and explanation of how they used ChatGPT and which strategy they applied during writing tasks.

- *Tailored learning and adaptive systems*: The model can be used in the design of interactive and adaptive AI platforms, where the online platform can interact metacognitively with the learner. The online platform can serve as a metacognitive detector by identifying the MR strategies students used or lacked. For instance, if the platform spots students using the use of same prompting skills, it can signal to the learner that they need to reconsider the output.
- *Cultivation of AI literacy and Autonomous learning*: The model can also be implemented to support novice or beginners in the use of AI tools in writing. Instructors can provide prompts and classify them based on each metacognitive strategy to develop metacognitive abilities. Regarding research, this model can also be of aid in the design of rubrics, scales, or observational tools to evaluate students' metacognitive abilities in using AI.

Primarily, the model can serve writing task designers and instructional practices. Writing instructors can use It as a heuristic for developing AI-integrated assignments, assisting learners to strategically plan, monitor, and evaluate their interaction with AI while maintaining authorship and academic integrity. Through incorporating these AI-specific metacognitive strategies, the model supports learners in elevating reflective, self-regulated approaches to AI-assisted writing.

Moreover, the model offers a contextualized framework for scaffolding in interventions. Educators can design activities that cultivate AI-related components, such as crafting effective prompts, authenticating information, and controlling hallucinations, along with the integration of reflection on writing processes. This approach advances metacognitive awareness and agency as core components to ensure that learners not only leverage AI tools effectively but also transfer these self-regulatory processes to other writing genres and tasks.

5 Conclusions and implications

The current study examined the use of metacognitive regulation (MR) strategies by Moroccan postgraduate students in employing ChatGPT for academic writing. The findings revealed nine main themes and seventeen sub-themes distributed across five dimensions of MR strategies. Key results indicated that students actively implemented diverse MR strategies to mitigate the limitations of ChatGPT while maximizing its benefits in essay writing. In other words, students with higher MR skills tended to use ChatGPT more productively. They also demonstrated positive attitudes and perceptions regarding the use of this technological tool for developing their writing skills.

This study makes three primary contributions to the existing literature. First, by adopting a qualitative research approach, it provides an in-depth and comprehensive understanding of MR strategies in the context of AI-assisted writing. This complements the limited research in the field and offers a reference point for developing metacognitive scales for future quantitative studies in AI-supported writing. Second, the research introduces a contextualized conceptual framework of MR specifically for GenAI-supported academic writing. Finally, the study addresses the lack of research at the national level and the limited studies internationally on the interaction between higher-order thinking skills particularly metacognition and GenAI technologies in academic writing contexts.

Given the rapid rise of GenAI technologies and their integration into educational plans and reforms, these tools are expected to play increasingly prominent roles in writing education. Accordingly, students should activate their metacognitive knowledge and regulation to effectively monitor their use of these powerful yet imperfect tools. This can be achieved through reflective teaching practices, such as think-aloud protocols or reflective reports, which promote self-regulatory judgment. Cultivating reflective thinking skills can lead to more efficient, strategic, and productive use of GenAI tools.

Based on these outcomes, several recommendations are offered for instructors, curriculum designers, and researchers. First, curriculum designers should incorporate tasks and activities aimed at enhancing students' AI literacy and promoting the effective use of these technologies across academic contexts. These activities may include familiarizing students with emerging AI tools, teaching effective prompting strategies, and raising awareness about their ethical implications. Second, instructors should leverage the benefits of GenAI technologies through focused training, such as workshops, webinars, and guided interventions. Within writing courses, educators are encouraged to explicitly train students on using AI tools to address writing challenges and improve writing proficiency.

In addition, students require explicit training in MR strategies. Instructors should teach students how to plan, monitor, and evaluate their use of GenAI tools in academic writing. Such training enables students to exercise cognitive control over AI-assisted writing, enhancing productivity and yielding positive outcomes. Therefore, educators and practitioners are encouraged to integrate emerging technologies into higher education, specifically by bridging metacognition and AI tools to support students' writing skills.

Finally, the model proposed in this study is intended for writing instructors, educators, and researchers. Writing instructors can implement AI-assisted writing training combined with MR strategies to enhance efficiency and productivity. This includes teaching students how to interact effectively with the tool—for example, using different types of prompts (e.g., role prompting, chain-of-thought prompts) and employing research strategies to gather relevant, focused ideas for writing. To strengthen monitoring and evaluation skills, instructors can expose learners to various ChatGPT-generated essay samples, collaboratively analyzing their strengths and weaknesses to enhance writing proficiency. Educators should also cultivate critical questioning of AI-generated content, ensuring that students do not follow outputs uncritically.

For researchers, the proposed model offers a foundation for further investigation, including quantitative and experimental studies. The framework serves as a guiding path for leveraging ChatGPT as a writing assistant, with MR and self-reflection at the forefront of promoting learner engagement and academic achievement.

6 Limitations and avenues for future research

The study primarily explored students' reported experiences with ChatGPT in the context of writing and metacognitive regulation (MR). Accordingly, it focused on participants' reflections and perceptions rather than observed task performance. This approach aligns with the interpretative nature of qualitative research and supports the development of a conceptual model grounded in lived experiences rather than measurable outcomes. The aim was not generalization, but the creation of a focused model rooted in experiential learning. Overall, the model underscores the role of MR in effective self-directed learning. Unlike many existing models of AI and metacognition, this framework emphasizes learner-initiated metacognitive engagement. It extends classical models of self-regulation (e.g., [61]) and metacognition [10, 18, 50] by contextualizing them within AI-assisted writing environments.

Despite its contributions, several limitations should be acknowledged. First, the study relied on self-report data and did not include observational or trace measures of AI-assisted writing. Second, the participant sample was highly homogenous, consisting exclusively of postgraduate English MA students with Advanced/C1 proficiency in English and at least two years of experience using AI tools. While purposive sampling enabled in-depth insights into expert users' metacognitive processes, it introduces selection bias. Consequently, the findings and resulting model may not generalize to undergraduates, novice AI users, or learners with lower language proficiency. Thus, the model should be interpreted as representing the MR strategies of experienced, high-proficiency L3 learners. It is also worth noting that member checking of transcripts or themes was not conducted due to time constraints and to minimize participant burden, particularly given the participants' engagement in intensive MA coursework.

Third, the study's scope was limited to ChatGPT-assisted writing. Nevertheless, the interactive model has potential applicability to other language modalities, such as reading, and is transferable to other AI applications. The deliberate focus on academic writing was intended to provide a conceptual foundation.

Regarding future research, studies may build on this contextualized conceptual model to explore these constructs using different methodologies or to develop new measurement scales across diverse AI environments. Researchers are encouraged to further investigate the interaction between AI and higher-order thinking skills including critical thinking, metacognition, and self-regulation using varied research approaches and designs. Within AI-assisted writing, experimental and longitudinal studies are particularly needed to assess the effects of different AI tools on psychological factors such as engagement, motivation, and anxiety across academic tasks.

Additional qualitative studies are recommended to uncover nuanced individual perspectives on students' interactions with these tools in other language-related skills, including speaking and reading, and in diverse educational contexts. Finally, continued investigation of the interplay among AI tools, higher-order thinking skills, and multilingual writing presents a fertile area for research, with the potential to yield valuable insights into the adoption and effective integration of AI in classrooms and academic programs.

Appendix A

Interview protocol:

Demographic part

1. How old are you ?
2. Is your mother tongue Moroccan Arabic ?
3. Do you use ChatGPT for academic writing ?
4. How long have been using AI ?

Planning

Can you describe how you prepare to use GenAI tools before starting an English academic writing task?

What factors influence your decision to use GenAI for specific aspects of your writing?

How do you set goals for integrating GenAI tools into your writing process?

Monitoring

Do you consciously and appropriately use ChatGPT during your writing tasks? If so, how?

While working on a writing task, how do you ensure that the output generated by GenAI aligns with your objectives?

Evaluating

After completing a writing task with GenAI, how do you evaluate the quality of the generated content?

Can you give an example of how you identified a mismatch between your expectations and the GenAI output?

What are the strengths of utilising ChatGPT for your writing tasks?

What are the drawbacks of utilising ChatGPT for your writing tasks?

Information management

When using ChatGPT, how did you use it to help you process, organise, elaborate and summarise the contents of your writing?

How do you ensure that the information generated is accurate, relevant, and suitable for academic purposes?

Debugging

What do you do when the GenAI output does not meet your expectations or contains errors?

How do you revise or refine prompts to achieve better results from GenAI tools?

Supplementary Information

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Supplementary Material 1.

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Author contributions

O.E. conceptualized the study, collected data, developed the methodology, and wrote the original draft. H.R. contributed to data curation, methodology, formal analysis, writing—review and editing, and validation. M.Y. contributed to the methodology and validation and participated in writing—review and editing. All authors reviewed the manuscript.

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Data availability

The pre-identified coding tables, theme definitions, and illustrative excerpts are provided in Supplementary File, while the 13-item interview protocol is included in the Appendix. Raw audio recordings or verbatim transcripts cannot be shared to ensure participant confidentiality.

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with accepted ethical standards for research. The present study received approval from our ethics committee at our institution of Chouaib Doukkali University in accordance with the ethical standards outlined in the 1964 Declaration of Helsinki and its later amendments. All participants voluntarily consented to participate in this study after being informed of the study's purpose and procedures.

Consent for publication
All participants in this study provided their informed consent to participate. They were informed that their responses in the interviews would be anonymised and used solely for research purposes, including publication in scientific journals. Their participation was voluntary, and they were assured that no identifying information would be disclosed at any stage of the research.

Informed consent
Informed consent was obtained from all participants prior to their participation in the study.

Competing interests
The authors declare no competing interests.

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References

1. Abdelhalim SM. Using ChatGPT to promoteresearch competency: English as a Foreign Language undergraduates' perceptions and practicesacross varied metacognitive awareness levels. J Comput Assist Learn. 2024;40(3):1261–75. https://doi.org/10.1111/jcal.12948
2. Abouabdelkader S. Moroccan EFL university students' composing skills in the balance: assessment procedures and outcomes. In: Springer eBooks. 2018. pp. 79–109. https://doi.org/10.1007/978-3-319-64104-1_4.
3. Alkamel MaA, Alwagieh NaS. Utilizing an adaptable artificial intelligence writing tool (ChatGPT) to enhance academic writing skills among Yemeni university EFL students. Social Sci Humanit Open. 2024;10:101095. https://doi.org/10.1016/j.ssaho.2024.101095.
4. Allen TJ, Mizumoto A. ChatGPT over my friends: Japanese English-as-a-Foreign-Language Learners' preferences for editing and proofreading strategies. RELC J. 2024. https://doi.org/10.1177/00336882241262533.
5. Albus P, Seufert T. Artificial intelligence in education: the importance of metacognitive monitoring. In: Studies in neuroscience, psychology and behavioral economics. 2024. pp. 109–18. https://doi.org/10.1007/978-3-031-70355-3_8.
6. Balalle H, Pannilage S. Reassessing academic integrity in the age of AI: A systematic literature review on AI and academic integrity. Social Sci Humanit Open. 2025;11:101299. https://doi.org/10.1016/j.ssaho.2025.101299.
7. Basu S, Dixit S. Role of metacognition in explaining decision-making styles: A study of knowledge about cognition and regulation of cognition. Pers Indiv Differ. 2022;185:111318. https://doi.org/10.1016/j.paid.2021.111318.
8. Belmekki NM, Belmekki NL, Chihab NM, Koumachi NB, Benmhamed NM. Writing in English and writing in Standard Arabic in Higher Education: a correlational study. Int J Engl Lang Stud. 2025;7(1):17–24. https://doi.org/10.32996/ijels.2025.7.1.3.
9. Braun V, Clarke V, Hayfield N, Davey L, Jenkinson E. Doing reflexive thematic analysis. In: Springer eBooks. 2022. pp. 19–38. https://doi.org/10.1007/978-3-031-13942-0_2.
10. Brown A. Metacognition, executive control, self-regulation and other more mysterious mechanisms. In: Weinert FE, Kluwe RH, editors. Metacognition, motivation and understanding. Hillsdale: Lawrence Erlbaum Associates; 1987. pp. 65–116.
11. Chen X, Li X, Zou D, Xie H, Wang FL. Metacognition research in education: topic modeling and bibliometrics. Education Tech Research Dev. 2025. https://doi.org/10.1007/s11423-025-10451-8.
12. Chkarka F, Fatmi H. Comparative examination of Master's students' and faculty members' maintenance of academic integrity in the age of AI. J Acad Ethics. 2026. https://doi.org/10.1007/s10805-025-09684-2.
13. Črček N, Patekar J. Writing with AI: University students' use of ChatGPT. J Lang Educ. 2023;9(4):128–38. https://doi.org/10.17323/jle.2023.17379.
14. Efklides A. Metacognition: Defining its facets and levels of functioning in relation to self-regulation and co-regulation. Eur Psychol. 2008;13(4):277–87.
15. Emran AQ, Mohammed MN, Saeed H, Keir MYA, Alani ZN, Ibrahim FM. Paraphrasing ChatGPT answers as a tool to enhance university students' academic writing skills. In: 2024 ASU International Conference in Emerging Technologies for Sustainability and Intelligent Systems (ICETSIS). 2024. pp. 501–5. https://doi.org/10.1109/icetsis61505.2024.10459386.
16. Errihani M. Language, power, and the economics of education in Morocco. In: Language policy. 2023. https://doi.org/10.1007/978-3-031-51594-1.
17. Ez-Zouhayry FE, Fallaki H. Towards an effective transformation of Moroccan higher education: a conceptual model for the digitalization of soft skills. In: Internationalization of higher education and digital transformation. 2025. pp. 349–69. https://doi.org/10.1007/978-3-031-76444-8_18.
18. Flavell JH. Metacognition and cognitive monitoring: A new area of cognitive–developmental inquiry. Am Psychol. 1979;34(10):906–11. https://doi.org/10.1037/0003-066x.34.10.906.
19. Flavell JH. Speculations about the nature and development of metacognition. In: Weinert FE, Kluwe RH, editors. Metacognition motivation, and understanding. New York: Lawrence Erlbaum Associates; 1987. pp. 21–9.
20. Gioia DA, Corley KG, Hamilton AL. Seeking qualitative rigor in inductive research: Notes on the Gioia methodology. Organizational Res Methods. 2013;16(1):15–31. https://doi.org/10.1177/1094428112452151.
21. Guamanga MH, Saiz C, Rivas SF, Bueno PM. Critical thinking and metacognition: Pathways to empathy and psychological well-being. J Intell. 2025;13(3):34. https://doi.org/10.3390/jintelligence13030034.
22. Guest G, Bunce A, Johnson L. How many interviews are enough? Field Methods. 2006;18(1):59–82. https://doi.org/10.1177/1525822x05279903.
23. Habib S, Vogel T, Anli X, Thorne E. How does generative artificial intelligence impact student creativity? J Creativity. 2024;34(1):100072. https://doi.org/10.1016/j.yjoc.2023.100072.
24. Han L. Metacognitive writing strategy instruction in the EFL context: focus on writing performance and motivation. SAGE Open. 2024. https://doi.org/10.1177/21582440241257081.
25. EL Hosayny O, Razkane H, Yeou M. Metacognition in academic writing: analyzing strategies and gender differences among Moroccan university EFL learners. J Arbiter. 2025;12(1):14–26. https://doi.org/10.25077/ar.12.1.14-26.2025.

26. EL Hosayny O, Razkane H, Yeou M. Extended TAM-based acceptance of AI for supporting CT in academic research: an examination of perceived benefits and challenges. Discov Artif Intell. 2025;5(1). https://doi.org/10.1007/s44163-025-00390-x.
27. Huang Y, Zhang K, Li S. How postgraduates navigate GenAI's dual pathways between research creativity and academic plagiarism: research engagement and self-regulated learning matter. Stud High Educ. 2025. https://doi.org/10.1080/03075079.2025.2543438.
28. Huang Y, Wu W, Wu P, Teng Y. Investigating the impact of integrating prompting strategies in CHATGPT on students learning achievement and cognitive load. In: Lecture notes in computer science. 2024. pp. 3–11. https://doi.org/10.1007/978-3-031-65881-5_1.
29. Junina AK, Strauss P, Wood JK, Grant L. A mixed-method inquiry into Arabic-speaking students' experiences with English academic writing at the undergraduate level. Asia Pac J Educ. 2025. https://doi.org/10.1080/02188791.2022.2101986.
30. Kaledio P, Robert A, Frank L. The impact of artificial intelligence on students' learning experience. SSRN Electron J. 2024. https://doi.org/10.2139/ssrn.4716747.
31. Kim J, Yu S, Detrick R, Li N. Exploring students' perspectives on Generative AI-assisted academic writing. Educ Inform Technol. 2024. https://doi.org/10.1007/s10639-024-12878-7.
32. Levine S, Beck SW, Mah C, Phalen L, PIttman J. How do students use ChatGPT as a writing support? J Adolesc Adult Lit. 2025. https://doi.org/10.1002/jaal.1373.
33. Li M. Non-native English-speaking (NNES) students' English academic writing experiences in higher education: A meta-ethnographic qualitative synthesis. J Engl Acad Purp. 2024;71:101430. https://doi.org/10.1016/j.jeap.2024.101430.
34. Loginov A. Digital metacognition: Awareness, regulation, and analysis of thinking in digital environments. Next Res. 2025;2(4):100770. https://doi.org/10.1016/j.nexres.2025.100770.
35. Mamad A, Vígh T. Moroccan EFL public university instructors' perceptions and self-reported writing practices. Heliyon. 2023;9(5):e15626. https://doi.org/10.1016/j.heliyon.2023.e15626.
36. Mazari N. Building metacognitive skills using AI tools to help higher education students reflect on their learning process. RHS-Revista Humanismo Y Sociedad. 2025. https://doi.org/10.22209/rhs.v13n1a04.
37. Mizumoto A. Data-driven Learning meets Generative AI: Introducing the framework of metacognitive resource use. Appl Corpus Linguistics. 2023;3(3):100074. https://doi.org/10.1016/j.acorp.2023.100074.
38. Monib WK, Qazi A, Mahmud MM. Exploring learners' experiences and perceptions of ChatGPT as a learning tool in higher education. Educ Inform Technol. 2025;30:917–39. https://doi.org/10.1007/s10639-024-13065-4.
39. Nugoroho A, Andriyanti E, Widodo P, Mutiaraningrum I. Students' appraisals post-ChatGPT use: Students' narrative after using ChatGPT for writing. Innovations Educ Teach Int. 2025;62(2):499–511. https://doi.org/10.1080/14703297.2024.2319184.
40. Oates A, Johnson D. ChatGPT in the classroom: Evaluating its role in fostering critical evaluation skills. Int J Artif Intell Educ. 2025. https://doi.org/10.1007/s40593-024-00452-8.
41. Ofem UJ, Owan VJ, Iyam MA, Udeh MI, Anake PM, Ovat SV. Students' perceptions, attitudes and utilisation of ChatGPT for academic dishonesty: Multigroup analyses via PLS–SEM. Educ Inform Technol. 2025. https://doi.org/10.1007/s10639-024-12850-5.
42. Oubadi Y. Exploring sources of errors in L3 English production of natives of Moroccan Arabic. Int J Lang Literary Stud. 2024;6(3):446–57. https://doi.org/10.36892/ijlls.v6i3.1859.
43. Oubibi M, Hryshayeva K, Huang R. Enhancing postgraduate digital academic writing proficiency: the interplay of artificial intelligence tools and ChatGPT. Interact Learn Environ. 2025. https://doi.org/10.1080/10494820.2025.2454445.
44. Oussou S, Kerouad S, Hdii S. The Relationship between EFL Students' Use of Writing Strategies and Their Writing Achievement. JURNAL ARBITRER. 2024;11(1):1–12. https://doi.org/10.25077/ar.11.1.1-12.2024.
45. Razkane H, Ben Hammou S, Sayeh AY. A constructivist inquiry into English-Medium Instruction in Moroccan higher education. Train Lang Cult. 2025;9(3):71–83. https://doi.org/10.22363/2521-442X-2025-9-3-71-83.
46. Razkane H, Sayeh AY, Diouny S, Yeou M. The impact of explicit metacognitive writing strategy training in English (L3) on Arabic-French–English trilingual Learners' writing outcomes. Engl Teach Learn. 2024. https://doi.org/10.1007/s42321-023-00156-7.
47. Sawi IA, Alaa A. Navigating the impact: a study of editors' and proofreaders' perceptions of AI tools in editing and proofreading. Discover Artif Intell. 2024. https://doi.org/10.1007/s44163-024-00116-5.
48. Sayeh AY, Razkane H. Historical accounts of the development of the multilingual situation in Morocco: contrasting monolingual to translanguaging approaches to education. SN Social Sci. 2022. https://doi.org/10.1007/s43545-022-00362-3.
49. Sayeh AY, Razkane H. Beyond monolingualism: the benefits of translanguaging in science education. SN Soc Sci. 2025. https://doi.org/10.1007/s43545-025-01152-3.
50. Schraw G, Dennison RS. Assessing metacognitive awareness. Contemp Educ Psychol. 1994;19(4):460–75. https://doi.org/10.1006/ceps.1994.1033.
51. Teng MF. Metacognition and autonomy in building a community for language learning through VR digital gaming. Computers Educ X Real. 2024;4:100060. https://doi.org/10.1016/j.cexr.2024.100060.
52. Teng MF. Metacognitive Awareness and EFL learners' perceptions and experiences in utilising ChatGPT for writing feedback. Eur J Educ. 2025a. https://doi.org/10.1111/ejed.12811.
53. Teng MF. Understanding EFL student writers' metacognitive awareness in utilizing ChatGPT. System. 2025. https://doi.org/10.1016/j.system.2025.103848.
54. Teng MF, Qin C, Wang C. Validation of metacognitive academic writing strategies and the predictive effects on academic writing performance in a foreign language context. Metacognition Learn. 2022;17(1):167–90. https://doi.org/10.1007/s11409-021-09278-4.
55. Volpe-White J. I know what I don't know: Metacognition in leadership learning. New Dir Student Leadersh. 2024;2024(183):121–30. https://doi.org/10.1002/yd.20633.
56. Wang S, Wang F, Zhu Z, Wang J, Tran T, Du Z. Artificial intelligence in education: A systematic literature review. Expert Syst Appl. 2024;252:124167. https://doi.org/10.1016/j.eswa.2024.124167.
57. Yao Y, Sun Y, Zhu S, Zhu X. A qualitative inquiry into metacognitive strategies of postgraduate students in employing ChatGPT for English academic writing. Eur J Educ. 2025. https://doi.org/10.1111/ejed.12824.

58. Yuan Y, Li H, Sawaengdist A. The impact of ChatGPT on learners in English academic writing: opportunities and challenges in education. Lang Learn High Educ. 2024;14(1):41–56. https://doi.org/10.1515/cercles-2023-0006.
59. Zhang J, Zhang LJ. The effect of feedback on metacognitive strategy use in EFL writing. Comput Assist Lang Learn. 2024;37(5–6):1198–223. https://doi.org/10.1080/09588221.2022.2069822.
60. Zhang D, Zhang LJ. Metacognition and self-regulated learning (SRL) in second/foreign language teaching. In: Springer international handbooks of education. 2019. pp. 883–97. https://doi.org/10.1007/978-3-030-02899-2_47.
61. Zimmerman BJ. Attaining self-regulation: a social cognitive perspective. In: Handbook of self-regulation. Academic Press; 2000. pp. 13–39.
62. Zou M, Huang L. The impact of ChatGPT on L2 writing and expected responses: Voice from doctoral students. Educ Inform Technol. 2024;29(11):13201–19. https://doi.org/10.1007/s10639-023-12397-x.

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